Shadow-Loom: An Experimental Framework for Combining Graphical Causal Reasoning with a Game-Engine-Style Graphical World Model of Narrative

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Abstract

Stories interest us not because they are sequences of likely next words, but because they have causes, secrets, and consequences. **Shadow-Loom** is an experimental open-source framework that takes a narrative and turns it into a typed, versioned *graphical world model*. Two engines act on that model, in the spirit of a game engine that pairs a physics simulator with a higher-level director. The first engine is a *causal physics* grounded in Pearl's ladder of causation (Pearl, 2009; Pearl and Mackenzie, 2018; Bareinboim et al., 2022) and the recently proposed counterfactual calculus over Ancestral Multi-World Networks (Correa and Bareinboim, 2025). The second is a *narrative physics* that scores the same graph against four structural reader-states — mystery, dramatic irony, suspense (Wilmot and Keller, 2020), and surprise (Itti and Baldi, 2009) — in the tradition of Sternberg's curiosity / suspense / surprise triad (Sternberg, 1992), and plays the role of the drama manager in classical interactive-narrative architectures (Mateas and Stern, 2003; Riedl and Bulitko, 2013; Ware and Young, 2014). Large language models are used only at the boundary: extraction, rendering, and audit. Identification, intervention, and counterfactual reasoning happen in typed code over the graph, much as a game engine restricts a stochastic agent to physically admissible actions (Ha and Schmidhuber, 2018; Park et al., 2023). The same loop — ingest, simulate, generate, audit, re-ingest — runs over user prose and produces a versioned world model with explicit, inspectable mathematical structure.

Two motivations drive the design. First, counterfactual reasoning sits exactly on the rung of Pearl's hierarchy where current LLMs are weakest. They are competent at observational and even interventional causal questions in text (Kıcıman et al., 2024), but they fail to act on the implicit beliefs and counterfactual states they themselves infer (Gu et al., 2026; Kim et al., 2023; Xu et al., 2025), and the foundations of rung-3 reasoning need dedicated graphical machinery (Correa and Bareinboim, 2025; Bareinboim et al., 2022; Halpern, 2016). Second, coding agents reliably outperform open-domain agents at fixed model capacity because their outputs can be compiled, executed, and tested against a typed substrate. Shadow-Loom gives narrative reasoning the same affordance: candidate continuations are propagated through the graph, checked against counterfactual-calculus-style independence and exclusion rules (Correa and Bareinboim, 2025), and re-ingested for round-trip verification (Madaan et al., 2023; Bai et al., 2022; Gu et al., 2024).

The system is presented as a research artefact, not as a benchmarked NLP model. Formal evaluation is left to future work. The more modest claim made here is that wiring a causal-graphical substrate underneath a constrained-LLM renderer is a tractable design point, with plausible relevance to controllable generation, narrative theory of mind, computational social science, and the digital humanities. Code, fixtures, and pipeline are released open source so that other researchers — in NLP, in computational social science, and in the digital humanities — can adapt the substrate to their own questions. Definitions and equations for each pipeline stage are given in the appendix.

1 Introduction

Most current LLM-driven story systems treat a narrative as a sequence-prediction problem and absorb the causal, temporal, and epistemic structure of the story into the weights of a single model. Recent work shows that such models are competent at *reading* causal relations from text (Kıcıman et al., 2024) but markedly weaker at *acting* on the implicit beliefs of characters they have just inferred (Gu et al., 2026; Kim et al., 2023; Cross et al., 2024); in unconditioned generation they tend to collapse onto a small set of shared plot tropes (Xu et al., 2025; Tian et al., 2024). The complementary direction explored here keeps an explicit, typed graph of the storyworld (Schank and Abelson, 1977; Mani, 2012; Elson, 2012) and performs identification, intervention, and counterfactual reasoning over it symbolically (Pearl, 2009; Halpern, 2016; Correa and Bareinboim, 2025), using the LLM only as a constrained renderer (Riedl and Young, 2010; Yao et al., 2019; Goldfarb-Tarrant et al., 2020; Yang et al., 2023). Concretely, a query like "what if Macbeth refuses to murder Duncan?" becomes a rung-2 do-operation on a typed event graph extracted from the play (App. B.2) rather than a free-form prompt to a language model; "does Poirot know the conspirators are still lovers?" becomes a finite belief-set membership test (App. B.4) rather than a single forward pass.

Motivation 1: counterfactuals are exactly where LLMs struggle. Pearl's hierarchy is provably non-collapsible. Rung-3 quantities are not in general identifiable from rung-2 data, let alone from rung-1 observation (Bareinboim et al., 2022). Empirically, LLMs reproduce surface causal patterns from their training corpus (Kıcıman et al., 2024) but degrade sharply on multi-hop belief-conditioned action prediction (Kim et al., 2023; Gu et al., 2026) and on story-internal counterfactual consistency (Xu et al., 2025). The questions that matter to a reader of a story — "what if Macbeth had refused?", "what if Roy had told Mary the truth?" — live on the rung where the modern foundations (AMWN + ctf-calculus, Correa and Bareinboim, 2025) are designed to operate, and where end-to-end neural generators are not. A typed graph plus an explicit do-operator and abduction step turns each such question into a finite sequence of well-defined operations (Pearl, 2009; Halpern and Pearl, 2005; Halpern, 2016) rather than into a free-form imagination task.

Motivation 2: coding agents work better because they compile and test. A familiar pattern in agent design is that coding agents outperform open-domain agents at fixed model capacity. Their outputs can be compiled, executed against a test suite, and rejected when they fail. The lever is the *typed substrate underneath the natural-language reasoning*, not the model itself. Shadow-Loom imports that lever into narrative. A continuation is rejected if (a) its causal propagation (Eq. 1) violates inertia or affordance gating; (b) its counterfactual branch fails an AMWN-style consistency, independence, or exclusion check (Correa and Bareinboim, 2025); or (c) its re-extraction does not match the brief that licensed it (the audit pass, in the LLM-as-judge tradition of Bai et al., 2022; Madaan et al., 2023; Gu et al., 2024). The graph plays the role of the compiler; the auditor plays the role of the test suite.

The framework combines four ideas which, to the best of my knowledge, have not been wired together in a single open-source pipeline:

1. A **Pydantic-typed world graph** (`WorldStateV1`) with entities, events, beliefs, locations, channels, and three kinds of edge (causal, social, spatial), every node and edge tagged with both a chronological *fabula* time and a narrative *syuzhet* index (Shklovsky, 1965; Genette, 1980).

2. A **causal physics engine** that implements Pearl's three rungs — observation, do-intervention, and abductive counterfactual — with a propagation rule gated by an explicit "Impact $>$ Inertia" threshold and spatial affordance checks.

3. A **narrative physics layer** that scores the same graph against closed-form information-state functionals for mystery, dramatic irony, suspense, and surprise. Suspense follows the hope/threat formulation of Wilmot and Keller (2020); surprise is binary KL divergence between a corpus-marginal prior and an evidence-updated posterior.

4. A **constrained-LLM renderer + recursive auditor** loop (Gu et al., 2024) that converts the simulation result into a typed creative brief, generates prose under that brief, and then re-extracts the prose to verify no "miracle steps" were introduced.

The system is named for the dual-graph metaphor: a *loom* of factual edges runs in parallel with one or more *shadow* counterfactual branches, in the sense of the AMWN construction of Correa and Bareinboim (2025). Section 2 sketches the pipeline; Section 4 states what is novel and what is borrowed; Section 5 gives the design rationale; Section 6 discusses relevance to NLP, reasoning, and social science; Section 7 states the limitations honestly. Equations and per-stage definitions are in App. A; an extended narrative walkthrough of the whole loop on a single fixture is in App. C; worked examples drawn from the sixteen bundled plot fixtures follow in App. B; and a separate appendix on the authoring user interface, with screenshots and step-by-step example sessions, is in App. D.

How to read this paper. The main text is intentionally short. It states the design, situates it in the literature, and records what is novel as a combination. Readers who want to understand the symbolic machinery in detail — the schema, the ego-graph extraction, the do-operator, the abductive counterfactual, the four affective scorers, the brief-and-audit loop — should treat App. A as the authoritative specification. App. C is a long-form description of the same pipeline expressed as a story: *Macbeth* is fed in as raw prose and followed step-by-step through ingestion, intervention, propagation, scoring, rendering, and audit, with every intermediate object shown. The eight example subappendices in App. B then show how each individual stage manifests on a different bundled fixture. App. D is for readers who want to understand or use the authoring workspace; it cross-links every UI surface to the pipeline stage it exposes and includes example sessions on real fixtures.

2 Pipeline overview

The pipeline is a five-phase loop over a versioned world model. Every version is ancestor-linked, so a counterfactual fork is literally a sibling row in a directed acyclic version tree, not a mutation of canon. Figure 1 sketches the loop.

```
prose
```
↓ ingestion (5-pass extraction)
```
WorldStateV1 (typed graph)
```
↓ ego-graph slice + AMWN sandbox
```
NetworkX MultiDiGraph
```
↓ causal physics (do, abduction, propagation)
```
candidate sandbox states
```
↓ narrative physics (affective scoring)
```
CreativeBrief (typed constraints)
```
↓ constrained LLM render
```
prose
```
↓ recursive audit + re-extraction
```
new VersionRow (factual or shadow)
```

Figure 1: The Shadow-Loom loop. The LLM is invoked only at the top (ingestion), the bottom (rendering), and the audit; everything in between is symbolic computation over the graph. See the appendix for definitions of each stage.

The five phases correspond to the five appendix subsections (A.2–A.7): **(1) Ingestion** runs five LLM agents to extract a typed `GlobalRegister`, then three concurrent specialist agents (physics, social, consequences) per chunk to extract events, relationships, channels, and entity-state deltas, with a programmatic validation and auto-repair loop. **(2) Causal physics** forks an Ancestral Multi-World Network sandbox, applies a do-operator for interventions, runs abduction for counterfactuals, and propagates effects through a topological sort of the causal sub-graph subject to inertia and affordance gating. **(3) Narrative physics / affective calculus** scores each candidate sandbox against four structural information-state functionals and six trait-trajectory emotional functionals. **(4) Generation** packages the highest-scoring candidate as a `CreativeBrief` of typed `ConstraintBlocks` and hands *only* the brief to a creative LLM. **(5) Audit and feedback** runs an LLM-as-judge over the prose, detects "miracle steps" (state changes with no licensing edge in the brief), and either accepts the version or re-runs generation with the auditor's feedback appended, up to a configurable retry budget.

The entire loop is exposed both through a NiceGUI authoring workspace (eight cross-linked tabs over a versioned project DAG; App. D) and through a Model-Context-Protocol server with roughly forty tools, so a downstream agent can drive shadow-branch creation, scoring, and promotion programmatically.

3 Theoretical lineage and computational counterparts

The system has been built theory-first: every named field in `WorldStateV1` corresponds to a specific narratological, cognitive, or causal-inference construct, and every field has a prior computational instantiation that the design either generalises or borrows from. The pairings the design depends on are summarised below; an extended mapping lives in the project's academic-foundations document.

Fabula vs syuzhet (Shklovsky, 1965; Tomashevsky, 1965; Genette, 1980; Bordwell, 1985; Bal, 2009). Two integer indices on every `EventNode`: chronological $f(e)$ and presentation $s(e)$. The four affective scorers reduce to set operations between the two projections (App. A.5).

Possible-worlds narratology (Ryan, 1991, 2001; Doležel, 1998; Pavel, 1986), modal-logical roots (Lewis, 1973), and counterfactual graphical models (Balke and Pearl, 1994; Shpitser and Pearl, 2008; Correa et al., 2021; Correa and Bareinboim, 2025). Every node and edge carries `world_id` $\in \{\text{factual}, \text{shadow}\}$; counterfactual queries fork a sibling `VersionRow`. The AMWN tag and three-rule mental model (consistency, independence, exclusion) come from Correa and Bareinboim (2025); I adopt the naming and architecture even though the typed narrative graph used here is not a rigorous SCM. AMWN supersedes the Twin-Network construction (Balke and Pearl, 1994) and the multi-network machinery of Shpitser and Pearl (2008), and is the counterfactual-completeness counterpart to do-calculus (Pearl, 1995, 2009; Bareinboim et al., 2022).

Greimas (Greimas, 1983), Propp (Propp, 1968), Bremond (Bremond, 1973), and the symbolic story-grammar tradition (Rumelhart, 1975; Mandler and Johnson, 1977; Kintsch and van Dijk, 1978; Schank and Abelson, 1977; Lehnert, 1981; Meehan, 1977; y Pérez and Sharples, 2001; Bringsjord and Ferrucci, 2000; Chambers and Jurafsky, 2008). `GlobalTrait` (`WORLD_`) realises Greimas's "Power" actant; `RelationshipEdge.power_dynamic` mirrors the sender / receiver / helper / opponent topology; `EventNode.event_type` $\in$ {choice, outcome, revelation, utterance} is a coarse generalisation of Propp's functions, closer in spirit to Bremond's triad. The plot-unit and script traditions prefigure both the typed-event graph (Schank and Abelson, 1977; Lehnert, 1981) and the goal-directed planner / generator pipeline (Meehan, 1977; y Pérez and Sharples, 2001); the unsupervised narrative event chains of Chambers and Jurafsky (2008) are a direct precursor to the typed `causal_topology` maintained at runtime.

Cognitive narratology (Herman, 2002; Fludernik, 1996; Zwaan and Radvansky, 1998; Gerrig, 1993) and constructionist comprehension (Trabasso and van den Broek, 1985; Graesser et al., 1994; Kintsch and van Dijk, 1978). The five fields of the situation-model (time, space, protagonist, causation, intention) are nearly the field set on `EventNode`; that readers materialise causal links explicitly during comprehension is the empirical justification for materialising them explicitly in storage.

Theory of mind (Premack and Woodruff, 1978; Wimmer and Perner, 1983; Baron-Cohen et al., 1985; Zunshine, 2006; Baker et al., 2009) and its modern LLM benchmarks (Kim et al., 2023; Gu et al., 2026; Cross et al., 2024). `Belief{value, confidence, inertia, established_at_fabula, acquired_via_event_id, acquired_via_channel_id}` is a discrete approximation of recursive ToM with provenance. The benchmarks above motivate the auditor's separate behaviour-given-belief consistency check.

Sternberg's curiosity / suspense / surprise triad (Sternberg, 1978, 1992; Brewer and Lichtenstein, 1982) and its computational realisations (Cheong and Young, 2015; Bae and Young, 2008; Wilmot and Keller, 2020, 2021; Wilmot, 2022). The four structural scorers (MYS, IRO, SUSP, SUR) extend the triad with dramatic irony (Booth, 1974; Gerrig, 1993). Suspense follows the hope/threat formulation of Wilmot and Keller (2020); surprise follows Itti and Baldi (2009) with a geometric prior update; emotional arc analytics owe to Reagan et al. (2016).

Information-theoretic and epistemic substrate (Shannon, 1948; Fagin et al., 1995; Alchourrón et al., 1985). `Channel` carries `intelligibility` $\in [0,1]$ per participant (generalising the legacy encryption boolean) and an `eavesdropped_by` relation derived from the `intelligibility_threshold`; belief revision uses an AGM-style `beliefs_added` / `beliefs_invalidated` delta.

Causal inference (Pearl, 1995, 2009; Pearl and Mackenzie, 2018; Spirtes et al., 2000; Bareinboim et al., 2022) and actual causality (Halpern and Pearl, 2005; Halpern, 2016; Woodward, 2003). Pearl's three rungs map directly to the query taxonomy used here (`ObservationQuery`, `InterventionQuery`, `CounterfactualQuery`); `mechanism` and `causal_force` on edges are motivated by Halpern's "actual cause"; ego-graph slicing is a heuristic Markov-blanket cut (Verma and Pearl, 1988). Recent benchmarking of LLMs as causal reasoners (Kıcıman et al., 2024) supports the hybrid stance taken here: LLMs propose edges (`extract_graph.py`); typed code identifies and propagates them (`causal_physics.py`).

Plan-and-render era of neural story generation (Riedl and Young, 2010; Jhala and Young, 2010; Martin et al., 2018; Yao et al., 2019; Goldfarb-Tarrant et al., 2020; Rashkin et al.,

2020; Yang et al., 2022, 2023) and its benchmarks (Mostafazadeh et al., 2016; Fan et al., 2018; Akoury et al., 2020; Tian et al., 2024; Xu et al., 2025). The `CreativeBrief` (Step 9) plus channel-aware draft (Step 10) is a typed, causal-graph-conditioned variant of the same plan-then-render pattern. The brief enumerates per-effect typed `ConstraintBlocks`; the auditor (Step 11) is the rescore / revise step (Goldfarb-Tarrant et al., 2020; Yang et al., 2022) specialised to narrative-causal rather than stylistic criteria, in the LLM-as-judge tradition (Zheng et al., 2023; Bai et al., 2022; Madaan et al., 2023; Gu et al., 2024).

Reading as simulation (Mar and Oatley, 2008; Oatley, 2016, 1995; Hogan, 2003). The reason a tight causal/affective model matters more than surface fluency: fiction is a cognitive simulator, and what is being simulated is a typed world.

Game engines, interactive narrative, and world models for AI (Mateas and Stern, 2003; Riedl and Bulitko, 2013; Ware and Young, 2014; Kreminski et al., 2020; Park et al., 2023; Ha and Schmidhuber, 2018; LeCun, 2022; Hao et al., 2023; Wang et al., 2024; Smelik et al., 2014; Summerville et al., 2018). Architecturally, Shadow-Loom is closer to a game engine than to a sequence model: a typed simulation step (causal physics) is wrapped by a director / drama-manager (narrative physics) and exposed to a stochastic policy (the LLM renderer) only through a constrained action interface, in the lineage of Mimesis / Façade-style interactive drama (Mateas and Stern, 2003; Riedl and Bulitko, 2013) and narrative planners such as CPOCL/Glaive (Ware and Young, 2014). The generative-agent line of work (Park et al., 2023) and the "world model" line in deep RL (Ha and Schmidhuber, 2018; LeCun, 2022; Hao et al., 2023) make the same bet from the opposite direction: that an explicit, manipulable latent world is what makes long-horizon coherent behaviour tractable. The shadow branches in Shadow-Loom play the role those world models play — a sandbox in which counterfactual rollouts can be evaluated before being committed — but with a typed, inspectable schema in place of an opaque latent state. The bundled example worlds and authoring tools relate Shadow-Loom to the broader procedural-content-generation tradition (Smelik et al., 2014; Summerville et al., 2018; Kreminski et al., 2020).

4 What is novel, what is borrowed

Borrowed. Every ingredient in Section 3 is borrowed: Pearl's ladder (Pearl, 2009; Bareinboim et al., 2022); the AMWN + ctf-calculus framework (Correa and Bareinboim, 2025); actual-causality semantics (Halpern, 2016); the suspense and surprise formalisms of Wilmot and Keller (2020) and Itti and Baldi (2009); the LLM-as-judge audit pattern (Bai et al., 2022; Madaan et al., 2023; Gu et al., 2024); the plan-and-render line in neural story generation (Riedl and Young, 2010; Yao et al., 2019; Goldfarb-Tarrant et al., 2020; Yang et al., 2023, 2022; Rashkin et al., 2020); the typed-event / story-grammar tradition (Schank and Abelson, 1977; Lehnert, 1981; Mani, 2012; Elson, 2012); and the narratological foundations (Shklovsky, 1965; Genette, 1980; Sternberg, 1992; Ryan, 1991; Herman, 2002; Zwaan and Radvansky, 1998).

Novel as a combination. I am not aware of another open-source pipeline that simultaneously: (a) runs all three Pearl rungs over a typed narrative graph; (b) tags every node and edge with an AMWN `world_id` (Correa and Bareinboim, 2025) so that counterfactuals are first-class persisted objects; (c) couples that machinery to closed-form scorers for the four classical reader-states of narrative cognition; (d) uses the resulting score as a loss function over candidate interventions, with the LLM constrained to render the winner; and (e) closes the loop by re-extracting the rendered prose and comparing it against the brief. Each ingredient is borrowed. The composition is the contribution: a typed two-engine substrate underneath a constrained renderer. Architecturally the closest analogue is a game engine for narrative: a typed simulation step gated by a drama-manager-like scorer (Mateas and Stern, 2003; Riedl and Bulitko, 2013; Ware and Young, 2014), driving a constrained stochastic actor in the spirit of recent generative-agent and world-model work (Park et al., 2023; Ha and Schmidhuber, 2018; Hao et al., 2023), but with the latent state replaced by an explicit causal-graphical schema.

Specific small constructions. The "Impact $>$ Inertia" propagation rule (Eq. 1) and the geometric prior update inside the surprise scorer (Eq. 5) are introduced here as deliberately simple choices that keep both physics monotonic in useful ways (App. A.5). The two-axis fabula/syuzhet sampling of the affective scorers (App. A.5) is, to my knowledge, also new.

5 Rationale for the design choices

Why a typed graph rather than free text + RAG? Recent stress tests show that LLMs can answer "what does X believe?" but fail to behave consistently with that belief in downstream action prediction (Gu et al., 2026; Kim et al., 2023). Belief and causal structure that are recoverable from prose are not, on present evidence, reliably *operated on* inside the prose-only loop. A typed graph makes belief, channel intelligibility, and causal mechanism inspectable and intervenable.

Why both a causal and a narrative physics? The causal physics decides *what is allowed to happen* given the graph; the narrative physics decides *which of the allowed things would be most readable*. Decoupling licensing from desirability lets the system score and rank candidate interventions before any text is generated, and lets the same engine serve both "what-if" analytical queries (rung 2/3) and high-level affective directives ("maximise dramatic irony for character X").

Why constrain the LLM rather than fine-tune one? The renderer's job is reduced to dialogue, description, and pacing within a mathematical envelope. Whatever the model's training distribution, the brief enumerates the causal edges, beliefs to preserve, channels to keep hidden, and trait shifts to honour. Errors that survive into prose are then catchable by the audit pass because they correspond to specific, testable claims — the "compile and test" analogue from Motivation 2.

Why AMWN tagging at the persistence layer? Treating a counterfactual as a sibling `VersionRow` with `world_id="shadow"` makes "what-if" branches first-class, diffable, promotable, and auditable; it is the persistence-layer analogue of the AMWN graphical construction (Correa and Bareinboim, 2025). A user can compare a shadow against canon, keep both, or promote.

6 Why this might matter beyond fiction

NLP / reasoning. Narrative is a tractable test bed for counterfactual and theory-of-mind reasoning because the ground truth (what happens, who knows what) is finite and authored. A typed-graph + symbolic-physics + constrained-renderer architecture is one concrete instantiation of the neuro-symbolic stance for controllable generation; the audit pass produces a falsifiable record of where the LLM departed from the brief. Recent ToM benchmarks (Gu et al., 2026; Kim et al., 2023; Cross et al., 2024) ask exactly the kind of behaviour-given-belief questions that an explicit `Belief` field with provenance is designed to support.

Computational social science and digital humanities. Many socio-historical and humanistic questions are counterfactual in form ("what if policy X had not been adopted?", "how would this novel read if the protagonist had spoken sooner?"). A graphical substrate that lets a researcher ingest a narrative source, fork a shadow branch under an explicit intervention, and inspect the propagated trait and relationship deltas is a small step toward auditable narrative simulation that can sit alongside established computational-social-science (Lazer et al., 2009) and digital-humanities tooling. I make no claim of empirical validity for any specific simulation; the contribution is the substrate, not its calibration.

Author tooling. The same machinery surfaces in a NiceGUI workspace and a Model-Context-Protocol server with around forty tools, exposing the graph, the affective scorers, and the version DAG to either human authors or LLM agents. The workspace is described component-by-component in App. D.

An invitation to other research communities. The substrate is deliberately domain-agnostic at the graph level: entities, events, beliefs, channels, and counterfactual branches are not specific to fiction. Computational social scientists (Lazer et al., 2009) routinely treat narrative sources — court records, parliamentary debates, oral histories, news streams — as data, and digital-humanities scholarship has long sought structured, sharable representations of literary works that can support both close and distant reading. Shadow-Loom can be read as one such representation: an open, typed, version-controlled graph with an explicit counterfactual sandbox and an audit trail back to the source text. I expressly invite researchers in those communities, as well as in narratology and cognitive science, to fork the schema, swap in their own ingestion prompts and affective scorers, and report what they find. The pipeline ships with sixteen worked example worlds drawn from canonical literature and film, which can serve as a common reference set for cross-group experimentation.

7 Limitations and future evaluation

Shadow-Loom is an experimental research artefact, not a benchmarked NLP system, and I am explicit about what that does and does not imply. I report no headline numbers against an external dataset: the scores produced by the narrative physics are design choices grounded in narratological theory rather than validated reader-reaction models, and the causal physics borrows the AMWN naming and three-rule mental model of Correa and Bareinboim (2025) without implementing their identification algorithm literally — the graph is heavily typed for narrative use rather than for rigorous structural causal model identification. The ingestion pipeline depends on LLM extraction, and although a programmatic validation and auto-repair loop catches the obvious failure modes, extraction errors can still propagate. The current unit-test suite covers the symbolic layers but does not constitute external evaluation.

Formal evaluation is therefore deferred to future work, along several complementary axes: (i) human-judgement studies of the affective scorers against reader annotations, in the spirit of Wilmot and Keller (2020); (ii) targeted ablations on bundled fixtures to quantify the contribution of the causal physics, the AMWN sandbox, and the audit loop to downstream coherence; and (iii) comparison against end-to-end LLM baselines on counterfactual and theory-of-mind benchmarks (Kim et al., 2023; Gu et al., 2026; Cross et al., 2024; Xu et al., 2025). I share the present design as a substrate for further research — by myself and by others — not as a finished product.

Code, licence, and reproducibility

The code is released open source under AGPL-3.0-or-later, with a parallel commercial licence; the authoritative LICENSE file lives in the project repository. The pipeline, schema, MCP surface, auditor, and the affective scorers described here all sit in the `shadow_loom/` package. Worked examples on bundled plots (Macbeth, Death on the Nile, Reservoir Dogs, and others) ship in `example_worlds/` and `sample_plots/`.

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A Definitions and equations

This appendix gives the formal definitions of the data model and the equations of each pipeline stage. Section references in the main paper point here.

A.1 World model schema (`WorldStateV1`)

A world state is a tuple

$$\mathcal{W} = (\mathcal{N}, \mathcal{E}, \mathcal{C}, \mathcal{T}, \tau)$$

with nodes $\mathcal{N}$ (entities ENT_•, events EVT_•, locations LOC_•, objects OBJ_•, channels CHN_•, world traits WORLD_•), edges $\mathcal{E}$ (causal, relationship, spatial), channels $\mathcal{C}$ with per-participant intelligibility $\iota \in [0, 1]$, world traits $\mathcal{T}$, and a branch tag $\tau \in \{\text{factual}, \text{shadow}\}$ inherited by every node and edge.

Each event $e \in \mathcal{N}_{\text{EVT}}$ carries two integer indices: chronological $f(e) \in \mathbb{Z}$ (*fabula time*) and presentation $s(e) \in \mathbb{Z}$ (*syuzhet index*), with s contiguous and unique. Trait values, beliefs, and ambient state are carried by typed vectors $(v, \iota_{\text{inertia}}, \sigma_{\text{evidence}})$ representing value, inertia, and evidence strength. Beliefs carry provenance (*event_id*, *channel_id*) and the time they became established. A pure function $\textsc{Reconstruct}(e_i, t)$ replays sparse `state_timeline` deltas to give the entity's effective state at any fabula time t.

A.2 Ingestion

Given prose P, ingestion produces a $\mathcal{W}$ via:

1. Five extraction agents over P build a `GlobalRegister` of WORLD_, ENT_, LOC_, OBJ_ nodes plus a fuzzy alias table.
2. For each chunk, a Socratic Who/What/Where/When/Why/How scaffold dispatches three concurrent specialist agents: *physics* (events, causal/spatial edges), *social* (relationship edges, channels), *consequences* (authoritative entity-state deltas).
3. Output is normalised, fabula times are remapped to a uniform $\Delta t = 100$ spacing, and a programmatic validator + LLM correction loop enforces ID closure and type constraints.

A.3 Ego-graph slicing and AMWN sandbox

Given a query with focal entities F and time anchor t, the ego-graph extractor returns

$$\mathcal{G}_{F,t} = \big(\textsc{Neighbours}_k(F) \cap \textsc{Reconstruct}(\cdot, t)\big),$$

the k-hop spatial / causal / social neighbourhood of F at fabula time t, with belief and trait values reconstructed at t so future information cannot leak into a counterfactual. The `AMWNInstantiator` mirrors $\mathcal{G}_{F,t}$ as a NetworkX `MultiDiGraph`, in the spirit of the AMWN construction (Correa and Bareinboim, 2025), with edge types {`located_in`, `owned_by`, `causal`, `relationship`, `connected_to`, `communicating_with`, `eavesdropped_by`}.

A.4 Causal physics

Given a sandbox $\mathcal{G}$ and an intervention specification:

Action (rung 2): $\text{do}(X = x)$ severs every incoming causal edge into X and writes $V_X \leftarrow x$.

Abduction (rung 3): for evidence E, each entity trait blends 50% toward its observed factual value, weighted by edge σ_{evidence} and gated by a typed `MECHANISM_TRAIT_MAP` that says which mechanisms can touch which trait families.

Propagation: a topological sort over the causal sub-graph applies, for each downstream node u with parent p and edge weight w:

$$I(u) = (V_p - V_u)\, w, \qquad \Delta_u = \begin{cases} I(u) - \operatorname{sgn}(I(u))\, \iota_u & \text{if } |I(u)| > \iota_u, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

A spatial affordance check additionally blocks cross-location influence unless an unbroken `connected_to` path exists. The result is a structured `CausalPhysicsResult{mutations, blocked, hidden_deltas, intervened_nodes}`.

A.5 Narrative physics (affective calculus)

Let A be the anchor pair (t_f, t_s) for fabula and syuzhet cursors, F the focal entities, and write *revealed* for any event e with $s(e) \le t_s$. Define $\mathrm{Anc}(e)$ as the causal ancestors of e in the graph and $B(F, t_f)$ as the union of belief sets of F reconstructed at t_f.

Mystery. For each revealed effect e,

$$\mathrm{Mys}(e) = \frac{|\{a \in \mathrm{Anc}(e) : s(a) > t_s\}|}{|\mathrm{Anc}(e)|}, \tag{2}$$

the fraction of e's causes still hidden from the reader.

Dramatic irony. For each revealed causal edge $(a \to e)$ in which a is not in $B(F, t_f)$, count one "irony gap"; the score is the fraction of revealed causal connections that are gaps.

Suspense (Wilmot and Keller, 2020). For each unrevealed event e with strongest incoming causal weight p_e:

$$w_{\text{threat}} = \sum_{e:\, F \subseteq \text{target}(e)} p_e, \quad w_{\text{hope}} = \sum_{e:\, F \cap \text{actor}(e) \neq \emptyset} p_e, \tag{3}$$

$$\mathrm{Susp} = \mathrm{clip}_{[0,1]}\left(\frac{w_{\text{threat}} - w_{\text{hope}}}{w_{\text{threat}} + w_{\text{hope}}}\right), \tag{4}$$

returning 0 when $w_{\text{hope}} = 0$ (the suspense–despair boundary).

Surprise. Per trait T, the prior q starts at the corpus marginal across entities (defaulting to 0.5), then for each revealed cause $(a \to e)$ with weight w it is updated by

$$q \mathrel{+}= w\, (\text{actual}_T - q), \tag{5}$$

a geometric pull that monotonically converges on the truth. Letting p be the actual trait value,

$$D_{\mathrm{KL}}(p \,\|\, q) = p \log \tfrac{p}{q} + (1 - p) \log \tfrac{1-p}{1-q}, \tag{6}$$

and the entity-level surprise is the mean over traits, normalised by $\log(1/\varepsilon)$ to $[0, 1]$ for a small ε.

Six emotion scorers (`grief, rage, joy, regret, love, fear`) declare which traits should move in which direction and score the headroom available, weighted by inertia evidence. The combined affective score requested by a `DirectiveQuery` is a weighted sum over the active functionals.

Two-axis sampling. The same scorers are evaluated along the syuzhet axis (advancing t_s only reveals more, so mystery, irony, surprise are non-increasing) and the fabula axis (snapshotting the world at t_f, where surprise can spike at intrinsically surprising story moments).

A.6 Generation

DirectiveAssembler.evaluate_candidate_events() forks the sandbox per candidate intervention, runs the causal physics, prunes candidates that violate inertia, affordance, or propagation constraints, ranks survivors by the affective score, and packages the winner as a CreativeBrief of typed ConstraintBlocks (per-effect): hidden-predecessor lists for mystery, "MUST NOT learn" guards for irony, threat/hope tables and hidden-channel lists for suspense, per-trait KL magnitudes for surprise, and per-trait shift constraints with headroom and inertia evidence for the emotion targets. Only the brief is shown to the renderer LLM, which produces dialogue, description, and pacing within the envelope.

A.7 Audit and feedback loop

The auditor (an LLM-as-judge in the sense of Gu et al., 2024) runs three passes over the rendered prose: (i) **causal** — reverse-engineers prose into causal claims and flags "miracle steps" (state changes with no licensing edge in the brief); (ii) **abduction** — runs counterfactual probes against the prose to check that implicit events hold; (iii) **affective** — measures the realised epistemic gap in the prose against the target. If the structured loss is non-zero, generation is re-run with the auditor's feedback appended, up to max_correction_retries. On success the re-extracted prose is merged into a new VersionRow, with world_id set by the branch policy (auto | mainline | shadow); counterfactual queries default to a fresh shadow branch, preserving canon.

B Worked examples on real plot fixtures

The repository ships sixteen hand-authored example_worlds/ fixtures paired with prose synopses in sample_plots/. A small subset is used below to illustrate each component end-to-end. Field values quoted below are verbatim from the fixtures; queries are the same ones run by the example walkthroughs in docs/model-examples.md and docs/pipeline-by-example.md.

B.1 World model — Macbeth

In example_worlds/macbeth.py, the location LOC_HEATH carries ambient_state{supernatural: AmbientVector(value=0.9, volatility=0.2, evidence_strength="strong"), concealment: 0.7}. The entity ENT_MACBETH is a bundle of typed TraitVectors: ambition, courage (high inertia 0.7, hard to shake), guilt (low inertia 0.25, highly mutable), and so on; an initial Belief(target=PROPHECY, confidence=0.4) sits on his belief list with its own inertia. OBJ_BLOODY_DAGGERS declares Affordance(action="kill", target_type="Entity") — the precondition gate the engine checks before it lets EVT_DUNCAN_MURDER fire. Every EventNode carries both indices: the murder has a fabula time but the syuzhet index follows the staged scene. The fixture exercises all five CausalEdge modalities (chain_reaction, mutation, mutation_social, affordance_gate, ambient_propagation); the ambient_propagation edge is what lets LOC_HEATH.supernatural=0.9 bias Macbeth's uptake of the prophecy belief.

B.2 AMWN sandbox — Macbeth "what if Macbeth refuses?"

The query "Macbeth refuses to murder Duncan" is a rung-2 intervention. AMWNInstantiator.create_sandbox(F={ENT_MACBETH}, t) mirrors the ego-graph as a MultiDiGraph; a sibling VersionRow with world_id="shadow" holds the fork. apply_do_operator({EVT_DUNCAN_MURDER: ∅}) severs the murder's incoming causal edges; the downstream chain_reaction edges into EVT_MALCOLM_FLEES, EVT_MACBETH_CROWNED, EVT_BANQUO_MURDERED all lose activation; the mutation edges into ENT_MACBETH.guilt and paranoia never fire. The factual mainline is untouched — the shadow row is browsable, diffable, and promotable from the UI.

B.3 Causal physics — Macbeth propagation under "Impact > Inertia"

After the do-operator, `propagate()` topologically sorts the causal sub-graph. For the original (factual) chain, the edge `EVT_BANQUO_GHOST` $\rightarrow$ `ENT_MACBETH.guilt` with weight $w = 0.7$ and source delta $V_p - V_u = 0.6$ produces $I = 0.42$. Because `guilt`'s post-shock inertia ι has been bumped to ~ 0.3 from its baseline 0.25 by the previous `state_timeline` entry, the gate $|I| > \iota$ passes and $\Delta = 0.42 - 0.3 = 0.12$ is written. A symmetrical attempt to shock `courage` (initial inertia 0.7) at $w = 0.5$, $\Delta_{\text{src}} = 0.4$ gives $I = 0.20 < 0.7$ and is *blocked* — exactly the play's psychology, in which his nerve outlasts his conscience. Spatial affordance gating additionally blocks cross-location shocks unless a `connected_to` path exists.

B.4 Narrative physics — Death on the Nile, Reservoir Dogs, Romeo and Juliet, Macbeth

The four structural scorers each have a canonical fixture.

Mystery (Macbeth Act II). Eq. 2 on the revealed effect `EVT_DUNCAN_FOUND_DEAD`: of its causal ancestors, `EVT_LADY_PERSUADES` and `EVT_DAGGERS_PLACED` are still hidden at the audience's `syuzhet_anchor` early in the act; the score is $2/3 \approx 0.67$, falling to 0 once Act V resolves the provenance.

Dramatic irony (Death on the Nile). The reader knows Jacqueline and Simon are conspirators (`EVT_PLAN_HATCHED` revealed at low syuzhet); Poirot does not. For each later event $(a \rightarrow e)$ where a is the conspiracy and $a \notin B(\text{ENT\_POIROT}, t_f)$, an irony gap is counted. The score sits high through the cruise scenes and only collapses at the denouement.

Suspense (Romeo and Juliet, tomb scene). At the tomb $F = \{\text{ENT\_ROMEO}\}$, unrevealed events with Romeo as non-acting target (Friar's letter intercepted, Juliet's draught about to wear off) sum to w_{threat}; events Romeo himself authors and that have not yet fired (drink poison, kill Paris) sum to w_{hope}. Threat dominates. As Romeo drinks, $w_{\text{hope}} \rightarrow 0$ and the score returns 0 — the Wilmot–Keller suspense–despair boundary is hit cleanly.

Surprise (Reservoir Dogs reveal). `EVT_ORANGE_RECRUITED` sits at low fabula time but high syuzhet; before the reveal the corpus-marginal prior on `ENT_ORANGE.role` starts at $q = 0.5$, and the few revealed edges into Orange give a small geometric pull keeping $q \approx 0.6$ "criminal". After the reveal the actual value $p \approx 0.95$ "cop". Eq. 5 gives a per-trait KL spike that is large but *licensed* — the engine had the cause present at low fabula time, only locked behind syuzhet, so re-extraction will not flag a miracle step.

Emotion targets (Gone Girl, Brief Encounter). The six trait-trajectory scorers (`grief, rage, joy, regret, love, fear`) declare which traits should move in which direction. On Gone Girl, the directive "maximise grief in Nick" resolves to required downshifts on `trust` and `hope` on `ENT_NICK`, weighted by inertia evidence; the available headroom score determines whether the directive is even feasible at the chosen anchor. On Brief Encounter the same machinery realises `regret` as Laura's $\Delta($`self_actualisation`$)$ headroom under a constant-supervision channel that never permits the choice to be made.

B.5 Directive assembly — Romeo and Juliet

The directive "raise dramatic irony to 0.85 in the tomb scene without giving Romeo correct information" triggers `DirectiveAssembler.evaluate_candidate_events()`. From unblocked affordances and reachable spatial paths it enumerates candidates — *Friar's letter intercepted*, *Balthasar arrives faster*, *Juliet wakes one minute earlier* — and forks an AMWN sandbox per candidate. The plausibility gate drops *Balthasar arrives faster* (the plague quarantine `SpatialEdge.is_locked` blocks the path); the remaining sandboxes are propagated, and the survivors are scored. *Friar's letter intercepted* wins because it raises Eq. 2's sister gap-count for irony without adding any utterance event into $B(\text{ENT\_ROMEO}, t_f)$. The winner is wrapped in typed `ConstraintBlocks`: a `must_events` list, a `must_not_events` list (Romeo learning the truth), trait envelopes (Romeo's `despair` envelope), and a syuzhet-window constraint. That bundle is the *only* thing handed to the renderer.

B.6 Generation — the brief as safety envelope

The renderer LLM is given the Romeo-and-Juliet brief as system prompt plus a small style context. It may write any prose about the tomb, the messenger, the dagger; it may not invent a causal edge, shift a trait outside its envelope, open a channel that does not exist, or place an utterance the brief does not licence. The output is a scene of dialogue, description, and pacing inside the mathematical envelope chosen by directive assembly — a typed, causal-graph-conditioned instance of the plan-and-render pattern (Riedl and Young, 2010; Yao et al., 2019; Goldfarb-Tarrant et al., 2020; Yang et al., 2023).

B.7 Audit and feedback loop — Gone Girl

Gone Girl is the system's lying-narrator fixture: Amy's diary is a sequence of `EventNodes` with `truth_value="false"` broadcast through `CHN_DIARY_PUBLIC`. After generation, `auditor.run_feedback_loop` runs three passes in parallel. The *causal audit* reverse-engineers the prose into $(\text{source}, \text{target}, \text{modality})$ claims and matches each against the brief; an LLM-emitted "Nick suddenly realises Amy is alive" with no licensing causal edge is flagged as a *miracle step*. The *abduction audit* runs counterfactual probes ("if the diary were truthful, would the prose still hold?"); a "yes" here means the prose has collapsed Amy's two faces into one, and is rejected. The *affective audit* measures the realised dramatic-irony gap against the target. The aggregated `FeedbackLoopResult` exposes both an `engine_thresholds_passed` gate (deterministic: number of miracle steps, max trait drift) and an `llm_pass` gate (literary critique); either failing re-runs generation with the audit feedback appended, up to `max_correction_retries`. On convergence the scene is re-extracted into a new `VersionRow`; counterfactual branches land on `world_id="shadow"` and only become canon via explicit `promote_branch` (Correa and Bareinboim, 2025).

C End-to-end walkthrough on *Macbeth*

This appendix describes a single complete pass of the pipeline in plain prose. The goal is to make every intermediate object inspectable to a reader who has read the main text but has not read the source. The fixture used throughout is `example_worlds/macbeth.py`, paired with the synopsis in `sample_plots/macbeth.txt`; the query is the canonical counterfactual "what if Macbeth refuses to murder Duncan?". The same loop runs identically on the other fifteen bundled fixtures.

C.1 Step 0: the source text and the user's query

The user opens the workspace, drops the synopsis into the STORY tab, and clicks *Save & Re-ingest*. The synopsis is a five-act prose summary in the public-domain register: the witches on the heath, Duncan's offer of the Cawdor title, Lady Macbeth's persuasion, the murder in Inverness, the banquet, the apparitions, the slaughter at Macduff's castle, the sleepwalking scene, Birnam Wood, and the final duel. After ingestion the user types a query into the REASONING tab: *"Counterfactual: Macbeth refuses to murder Duncan at Inverness. Show what changes for Macduff, Banquo, Malcolm, and the witches, and re-render the banquet scene."* The query is parsed by `query_parsing.parse_query` into a typed `CounterfactualQuery` with focal entities $F = \{\text{ENT\_MACBETH}\}$, target $X = \text{EVT\_DUNCAN\_MURDER}$, intervention value $x = \oslash$ (do-not-occur), and a re-render directive scoped to the banquet's syuzhet window.

C.2 Step 1: ingestion in detail

Ingestion is the only stage where the LLM is allowed to author graph structure rather than prose. It is a fan-out / fan-in pattern over the chunked source.

Ontology pre-pass. Five extraction agents run over the full text in parallel: `ontology_locations` produces `LOC_HEATH`, `LOC_INVERNESS_CASTLE`, `LOC_FORRES_COURT`, and so on, each with an initial `ambient_state` dictionary; `ontology_objects` extracts `OBJ_BLOODY_DAGGERS`, OBJ_LETTER, and OBJ_CROWN with explicit Affordance lists; ontology_entities produces ENT_MACBETH, ENT_LADY_MACBETH, ENT_DUNCAN, and the rest, each as an Entity with a sparse initial TraitVector bundle, beliefs, and a first-appearance location_id; ontology_world_traits introduces the world-level named latents (WORLD_AMBITION, WORLD_FATE, WORLD_GUILT) that act as common causes; and ontology_aliases builds a fuzzy alias table ("Lord of Glamis" $\mapsto$ ENT_MACBETH). The result is a populated GlobalRegister.

Per-chunk specialist fan-out. The text is chunked into ~ 800-token windows aligned on paragraph boundaries. Each chunk dispatches three concurrent specialist agents that share the GlobalRegister as system context: physics_extraction produces EventNodes with event_type drawn from {choice, outcome, revelation, utterance}, causal edges with causality_type, and spatial edges (e.g. a connected_to edge from LOC_INVERNESS_CASTLE to LOC_FORRES_COURT); social_extraction produces RelationshipEdges with per-axis RelationshipMetrics for affinity, fear, and power dynamic, and the Channels (the witches' prophecy channel, the dinner-table channel at the banquet, the diary-style private channel of Lady Macbeth's letter); and consequences_extraction writes the authoritative state_timeline entries on each entity, including the inertia bumps that record how shocking events make traits *harder* to dislodge afterwards (Macbeth's guilt inertia rises from 0.25 to 0.4 after the murder).

Validation and auto-repair. The merged graph passes through _programmatic_validation, which checks ID closure (every source_id resolves), type constraints (every causality_type is one of the five enums), basic acyclicity within causal edges of the same fabula slice, and "alive-actor" constraints (no event sourced from a dead character). Failures are passed back to a small repair LLM with the offending field and a structured error; the repair loop runs up to three iterations before raising. Fabula times are then remapped to a uniform $\Delta t = 100$ spacing so that downstream windows are stable across re-ingestion. The output is a single WorldStateV1, persisted as the root VersionRow.

C.3 Step 2: query routing and ego-graph slicing

The CounterfactualQuery is dispatched to the rung-3 handler. The handler first consults the NarrativePhysics router, which selects an *ego-graph* around the focal entities at the query's fabula anchor: Macbeth's k-hop neighbourhood at $t \approx 6000$ (the Inverness night). The slice contains ENT_MACBETH, ENT_LADY_MACBETH, ENT_DUNCAN, the chamber attendants, OBJ_BLOODY_DAGGERS, LOC_INVERNESS_CASTLE, the relevant beliefs, and the immediate causal context (EVT_LADY_PERSUADES, the prophecy events, the upcoming EVT_DUNCAN_MURDER). Trait values are reconstructed at t by replaying state_timeline deltas, which prevents post-murder trait shocks from leaking into the counterfactual past.

C.4 Step 3: AMWN sandbox and the do-operator

AMWNInstantiator.create_sandbox(F={ENT_MACBETH}, t) mirrors the ego-graph as a NetworkX MultiDiGraph, in the spirit of the Ancestral Multi-World Network construction of Correa and Bareinboim (2025). A sibling VersionRow with world_id="shadow" is opened to hold the fork. The do-operator apply_do_operator({EVT_DUNCAN_MURDER: $\oslash$}) severs the murder's incoming causal edges (specifically the chain_reaction edge from EVT_LADY_PERSUADES, the affordance_gate edge from OBJ_BLOODY_DAGGERS, and the ambient_propagation edge from LOC_INVERNESS_CASTLE.tension) and overrides its realisation to "not-occurred". Crucially the factual mainline is untouched. A user looking at the version sidebar sees a green factual row with a violet shadow child, both browsable.

C.5 Step 4: causal physics propagation

The propagator topologically sorts the causal sub-graph from the intervened node downward and applies Eq. 1 at every edge. Concretely:

- `EVT_DUNCAN_MURDER` $\rightarrow$ `EVT_MALCOLM_FLEES` (`chain_reaction`, $w = 0.85$): the source's outcome is "$\emptyset$", so the impact is zero, and `EVT_MALCOLM_FLEES` is marked `blocked_by_intervention`.

- `EVT_DUNCAN_MURDER` $\rightarrow$ `ENT_MACBETH.guilt` (`mutation`, $w = 0.7$): blocked. The downstream `state_timeline` delta that would have raised guilt from 0.1 to 0.7 is suppressed; the entity's reconstructed trait at the banquet remains close to its battlefield baseline.

- `EVT_DUNCAN_MURDER` $\rightarrow$ `EVT_BANQUO_SUSPECTS` (`chain_reaction_via_belief`, $w = 0.6$): blocked. `ENT_BANQUO`'s suspicion belief never fires; the `Belief(target=ENT_MACBETH, perceived_state= "Macbeth might be the king's killer")` is absent from his reconstructed belief set in the shadow branch.

- `EVT_LADY_PERSUADES` $\rightarrow$ `ENT_LADY_MACBETH.dominance` (`mutation_social`, $w = 0.5$): *not* blocked, because the persuasion event itself is upstream of the intervention. Lady Macbeth's dominance still spikes; her subsequent collapse arc is then re-evaluated against the now-absent causal trigger of guilt.

- Spatial affordance check at every step: a putative shock from `LOC_DUNSINANE_CASTLE` to `ENT_MACDUFF` requires a connected spatial path; if Macduff is in `LOC_ENGLAND` with no open channel, the shock is blocked.

The result is a structured `CausalPhysicsResult{mutations: [...], blocked: [...], hidden_deltas: [...], intervened_nodes: [EVT_DUNCAN_MURDER]}`. The `blocked` list is what makes the engine inspectable: every causal edge that *would have* fired in the factual mainline is recorded with the reason it did not.

C.6 Step 5: narrative-physics scoring

The four structural scorers and the six emotion scorers are now re-evaluated on the shadow branch. The user's directive ("show what changes for Macduff, Banquo, Malcolm, and the witches") biases the dramatic-irony scorer toward those four entities. The counterfactual collapses several gaps that defined the factual mainline: the audience no longer sees Banquo's suspicion gap, no longer sees Macduff's grief-driven vengeance, and the apparition scene is now etiologically dangling (the witches still prophesy, but Macbeth has no kingship to defend). The mystery score on `EVT_DUNCAN_FOUND_DEAD` drops to zero because the event no longer fires; the suspense at the banquet resets, since Banquo's ghost depended on the murder. The directive assembler records this delta as a `CandidateScoreReport` per candidate continuation it considers for the banquet rewrite.

C.7 Step 6: directive assembly and the brief

`DirectiveAssembler.evaluate_candidate_events()` now has to choose what to put in the banquet scene of the shadow branch. Candidate events are enumerated by walking unblocked affordances reachable from the shadow-branch state: *Macbeth confesses the prophecy to Banquo*, *Lady Macbeth persists and stages the murder herself*, *Duncan publicly names Macbeth as the new Prince of Cumberland*, *Macbeth retreats to Glamis to think*. Each is forked into its own micro-sandbox and propagated. Plausibility gates drop *Lady Macbeth stages the murder herself* because the daggers' `Affordance(action="kill")` is gated on a wielder with sufficient `ruthlessness` and the trait-evidence support is low for Lady Macbeth in isolation. The remaining survivors are scored against the directive. The winner is wrapped as a `CreativeBrief`: a list of `must_events`, a list of `must_not_events` ("Macbeth kills Duncan", "Banquo dies"), per-trait envelopes (Macbeth's `guilt` envelope is now $[0.1, 0.3]$ rather than $[0.6, 0.8]$), and a syuzhet window. The brief is the only object handed to the renderer.

C.8 Step 7: constrained rendering

The renderer LLM is invoked with the brief as system prompt plus a small style context (the synopsis's voice settings: synoptic-narration, third-person past tense, sparse density). It writes the banquet as a tense scene of overheard whispers, of Banquo unsuspecting and gracious, of Macbeth conspicuously unable to enjoy his elevation; the daggers stay sheathed; no ghost appears, because Banquo is alive. The renderer is free to choose dialogue, description, and pacing, but cannot invent a causal edge, shift a trait outside its envelope, or open a channel that does not exist. The output is a prose passage, not a graph.

C.9 Step 8: audit, re-extraction, and merge

The auditor runs three passes in parallel. The *causal audit* reverse-engineers the prose into (source, target, modality) triples and matches each against the brief; an LLM-emitted "Banquo's ghost rises behind Macbeth" would be flagged as a *miracle step* (no licensing edge), but in this run nothing of the kind appears. The *abduction audit* runs counterfactual probes against the prose — "if Banquo had suspected the prophecy, would the prose still hold?" — and checks the answers against the brief's belief envelope. The *affective audit* re-runs the four scorers on the prose and compares them to the brief's targets. If any pass fails the `engine_thresholds` gate or the `llm_pass` gate, the renderer is re-invoked with the structured feedback appended, up to `max_correction_retries` (default 3). On convergence, `ingestion` is run again on the prose alone, and the resulting graph is diffed against the brief; the diff becomes a new `VersionRow` with `world_id="shadow"` and `ancestor_id` pointing at the original counterfactual root. The user sees the new version appear in the sidebar in violet, can diff it against the factual mainline, and can `promote_branch` it to canon if desired.

C.10 What this walkthrough demonstrates

Three properties matter. **Inspectability:** every intermediate object — the ego-graph, the `CausalPhysicsResult`, the `CreativeBrief`, the `AuditReport` — is a typed Pydantic value, persisted and visible in the UI. **Locality of LLM authority:** the LLM authors the initial ontology (Step 1), the candidate prose (Step 7), and the audit critique (Step 8); everything else is symbolic computation over typed code. **Branch safety:** the counterfactual is a sibling row, not a mutation; canon is preserved, the shadow is diffable and promotable, and the version DAG records the lineage.

D The authoring user interface

The authoring workspace is a NiceGUI single-page application (`shadow_loom_ui/app.py`) reachable on `http://localhost:7860`. It is organised as a left-hand *version sidebar* and eight cross-linked tabs that share a central `AppState` pub/sub event bus (`shadow_loom_ui/state.py`). The same active-version pointer is mirrored to the database so an MCP client and the UI always see the same tip. This appendix walks through every surface in turn, with a concrete example session drawn from the bundled plot fixtures.

D.1 Version sidebar

The sidebar (`shadow_loom_ui/components/version_sidebar.py`) renders the project's full `VersionRow` DAG as a tree rooted at the original ingestion. Factual rows are green, shadow rows (counterfactual forks under `branch_policy=auto`) are violet. Toolbar actions: *Delete* re-parents children through `db.delete_version`; *Reparent* changes a row's `ancestor_id` (rejected if it would multi-root the tree); *Promote to canon* (shadow rows only) calls `db.promote_branch` to copy a shadow onto a fresh factual row; *Diff against factual* (shadow rows only) opens a structured diff against the current factual head. Clicking any row swaps the active version and resets every tab.

Example. After running the *Macbeth refuses the murder* counterfactual of App. C, the sidebar shows a green chain `ingestion` → `manual_edit` → `pipeline_run` for the factual mainline, with a single violet child branching off the `pipeline_run` row. Right-clicking the violet row offers *Diff against factual* (showing the absent `EVT_DUNCAN_MURDER` and the suppressed downstream deltas) and *Promote to canon* (which would replace the mainline).

D.2 Story tab

The STORY tab (`components/story_tab.py`) is the entry point: a plain-prose textarea plus a *Save & Re-ingest* button. Saving runs the full `pipeline.run_pipeline` and persists the result as a new version with `source="ingestion"`. This tab is intentionally not autosaved: the textarea is prose, not graph, and clicking re-ingest is a deliberate action.

Example. Pasting the public-domain `sample_plots/death_on_the_nile.txt` synopsis and clicking *Save & Re-ingest* produces, after the ingestion fan-out described in App. C.2, a `WorldStateV1` containing `ENT_POIROT`, `ENT_LINNET`, `ENT_JACQUELINE`, `ENT_SIMON`, the `LOC_KARNAK` steamer, and the conspiracy event `EVT_PLAN_HATCHED` with `syuzhet_index` placed late but `fabula_time` placed early — the very gap that the dramatic-irony scorer reads off in App. B.4.

D.3 Explorer tab

The EXPLORER tab (`components/explorer_tab.py`) is a read-only tree view of every node in the active world model, grouped by category, with an inspector panel for the selected node. It is the canonical way to confirm that ingestion has produced the right schema before running queries.

Example. Loading the Macbeth fixture and clicking `ENT_MACBETH` opens an inspector showing each `TraitVector` (`ambition: 0.7, inertia 0.55, evidence strong`; `guilt: 0.1, inertia 0.25, evidence weak`), each `Belief` with provenance, the `state_timeline` entries (one per major shock event), and the entity's outgoing causal and relationship edges. The inspector is the principal debugging tool when an extraction agent has produced an unexpected trait or belief.

D.4 World tab

The WORLD tab (`components/world_tab.py`) is a Cytoscape graph visualisation of entities, locations, objects, and edges, with a fabula-time slider beneath. Sliding the cursor through fabula time updates node colours and sizes via `state.set_fabula_cursor` (debounced 120 ms) so that the viewer can watch the social topology evolve.

Example. On Reservoir Dogs, scrubbing the slider from fabula $t = 0$ (the diner conversation) through to $t \approx 8000$ (the warehouse stand-off) animates the `ENT_ORANGE` node turning from neutral grey to red as the wound state advances; the spatial graph contracts as characters converge on the warehouse; and the `trust` edges from `ENT_WHITE` to `ENT_ORANGE` thicken. The same scrub on Romeo and Juliet shows Romeo's `despair` edge widening sharply over the final scene.

D.5 Causality tab

The CAUSALITY tab (`components/causality_tab.py`) has three sub-tabs. *Topology* renders a Sankey diagram of causal flow up to the fabula cursor: thicker ribbons for higher-weight edges, colour banded by `causality_type`. *Evolution* plots trait trajectories per entity over fabula time, with a vertical needle synced to the cursor. *Affective Dashboard* shows gauges and time series for the four structural scorers and the six emotion scorers, computed by `viz_helpers.compute_affective_scores` (which routes through `DirectiveAssembler.compute_*_score`). The top twenty entities by event degree are shown by default to keep the plots readable.

Example: Macbeth Evolution sub-tab. Selecting `ENT_MACBETH`, `ENT_LADY_MACBETH`, and `ENT_MACDUFF` and scrubbing through fabula time shows Macbeth's `guilt` curve climbing in two steps (after `EVT_DUNCAN_MURDER` and again after `EVT_BANQUO_GHOST`); Lady Macbeth's `ruthlessness` curve peaks early and then collapses into the sleepwalking scene; and Macduff's `grief` curve has a single step at the slaughter event. Switching to the *Affective Dashboard*, the suspense gauge sits high through Acts II–IV and zeroes out cleanly when the duel begins (Wilmot and Keller, 2020's suspense–despair boundary).

Example: Death on the Nile Affective Dashboard. The dramatic-irony gauge sits at ~ 0.7 from the moment the conspiracy is revealed to the audience and drops to zero only when Poirot delivers the denouement; the mystery gauge follows the inverse trajectory, peaking on the morning of the murder.

D.6 Reasoning tab

The REASONING tab (`components/reasoning_tab.py`) is where queries are typed and where intervention / counterfactual traces are inspected. The chat box accepts natural-language queries that are parsed into one of the eight typed query objects (`ObservationQuery`, `InterventionQuery`, `CounterfactualQuery`, `DirectiveQuery`, `InterrogateQuery`, `GeneralQuery`, `ManualEditQuery`, `EvaluateQuery`). The result panel uses `reasoning_helpers.py` to format the `CausalPhysicsResult`: each mutation is shown with its (I, ι, Δ) triple, each blocked propagation with the reason (inertia, affordance, spatial), each abduction back-fill with the evidence weight, and any hidden deltas with the syuzhet window in which they will be revealed.

Example. On Macbeth, typing *"If Macbeth had told Banquo about the prophecy on the heath, would Banquo have trusted him at the banquet?"* parses to a `CounterfactualQuery` with focal entities {ENT_MACBETH, ENT_BANQUO}. The trace panel shows the abduction step (Banquo's `trust` prior is inferred from the battlefield events), the do-step (a new `utterance` event is inserted on the heath channel), and the propagation step (Banquo's `suspicion` trait is suppressed at the banquet, but his `prudence` trait *raises* a new caution gate, which the engine reports as a non-trivial counterfactual outcome — not all secrets shared are secrets defused).

D.7 Audit tab

The AUDIT tab (`components/audit_tab.py`) displays the structured `AuditReport` for the most recently rendered scene. The three sections (causal, abductive, affective) each show their loss values and any offending prose spans, highlighted inline. A green badge indicates a passed pass; an amber badge indicates a warning (the prose is acceptable but drifted); a red badge indicates a hard failure that triggered a re-render.

Example. On Gone Girl, after generating a continuation of Amy's diary entry, the causal audit flags a hallucinated "Nick suddenly suspects Amy is alive" span with no licensing causal edge in the brief. The auditor's structured feedback is displayed alongside the offending span, and the user can either accept the auto-rerun or open the brief in the REASONING tab to widen the envelope.

D.8 Editor tab

The EDITOR tab (`components/editor_tab.py`) is the manual world-model editor. It has three integrated surfaces backed by a single JSON textarea (the source of truth): a *Validate* button that runs the same `_programmatic_validation` as ingestion; a *Structural editor* with one collapsible panel per collection (locations, entities, objects, events, world traits, causal edges, relationship edges, spatial edges, channels); and the raw JSON textarea itself for fields the structural editor does not expose (sparse trait values, beliefs, ambient state, evidence strengths). The save flow runs Pydantic validation, then `_programmatic_validation`; errors block the save with a dialog listing every issue, warnings present a *Save Anyway* confirmation, and on confirm the world is persisted as a new `VersionRow` with `source="manual_edit"` parented to the current row.

Example. Authoring the `example_worlds/macbeth.py` fixture from scratch, the user adds `LOC_HEATH` via the structural editor (typing *"the heath"* into the ID field auto-prefixes to `LOC_THE_HEATH`, edited to `LOC_HEATH`), then `ENT_MACBETH`, then the witches as a single multi-actor entity, then the prophecy event with `event_type= revelation` and `via_channel_id` pointing at a freshly added `Channel(participants=[ENT_MACBETH, ENT_BANQUO, ENT_WITCHES], medium="speech", intelligibility={...: 1.0})`. After each addition the validator panel updates: the green panel shows the running counts, an amber panel warns about the missing mutation coverage on ENT_MACBETH.ambition (no state_timeline yet), and a red panel reports the broken addressee_ids reference until the witches entity is added.

D.9 Export tab

The EXPORT tab (components/export_tab.py) is the egress surface: download the active world as a single world_state.json, download the version tree as a structured JSONL file, or copy the active version row ID for use with the MCP server or scripts.

Example. A researcher running cross-fixture comparisons exports each of the sixteen bundled fixtures' world_state.json files, computes the entity-degree distribution and the per-fixture suspense time-series in a notebook, and persists the results without ever loading the UI's visualisation code.

D.10 Performance discipline

Three small disciplines keep the workspace responsive during long scrubs and rapid edits. (i) Cursor writes are debounced through state.set_fabula_cursor, which schedules an emit on a 120 ms tail. (ii) Heavy panels are gated by is_path_visible(path) so that off-tab rebuilds are skipped. (iii) state.spawn_panel_task(panel_id, coro) cancels any prior task with the same panel_id, collapsing rapid scrubs to the most recent render; ECharts panels build the chart once and patch options in place.

D.11 Cross-tab event bus

Six events flow through AppState: PROJECT_LOADED (emitted by the project picker and the MCP open_project tool, listened to by every tab); WORLD_STATE_CHANGED (emitted by load_db_version, manual save, and pipeline runs, triggering cache invalidation in viz_helpers); VERSION_CHANGED (sidebar highlight + raw-text reload); FABULA_CURSOR_CHANGED and SYUZHET_CURSOR_CHANGED (cursor sliders); and ACTIVE_PATH_CHANGED (top tab change, gating heavy panels). Adding a new tab is therefore a matter of subscribing to the relevant subset of these events and rendering against the current AppState.

D.12 Why a UI matters for a research artefact

The UI is not an optional consumer surface; it is the principal debugging tool for the symbolic layers. The version sidebar makes branch lineage visible; the Causality tab makes the affective scorers' trajectories visible; the Reasoning tab makes the causal-physics trace visible; and the Audit tab makes the brief-vs-prose mismatch visible. A reviewer or collaborator can load any of the sixteen bundled fixtures, run the counterfactuals discussed in App. B, and inspect every intermediate object without writing a line of code. That inspectability is what the typed-substrate design was meant to buy; the UI is where it is cashed in.